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CleanStart CMS — Build Backlog

Phase A–I sequence for [apps/cms](#) (Payload admin + REST API + hooks). Each ticket links to an anchor in [docs/cleanstart-cms-architecture.html](#) for the *what*; conventions live in [CLAUDE.md](#) for the *how*.

Hard rule: a phase does not start until the previous phase's exit criteria are signed off.

Phase status at a glance

Phase	Status	Commit range
A · Bootstrap	✔ Done	c1317c7
B · Core schema	✔ Done	3d2298e–c27c735
C · Page-builder blocks	✔ Done	e19ce1c–0f04f87
D · Editor experience (server-side validators)	🟡 Partial	4665232–2c748b7
E · Forms + leads runtime	—	—
F · Search + structured data	—	—
G · Webhooks, cron, observability	—	—
H · Migration ETL	—	—
I · Hardening	—	—

Phase B verified end-to-end: DB drop+recreate, full schema push for all 17 collections + 6 globals, admin login, dashboard render with all groups visible.

Phase C verified end-to-end: 18 blocks render in the Pages "Add Layout" picker (Section primitive + 17 content blocks), Section composes nested blocks (one-level nesting only by design), full schema push clean.

Phase D **server-side** done in this session: block-level minRows validators (FAQ items, Pricing tiers, Section children, Table headers/rows, etc.), Lexical body-stats hook (readingMinutes/wordCount/tableOfContents on Blogs/News/Guides — 220 wpm, H2-H6 with disambiguated anchors), URL-shape and target validators on the typed-link field. **Client-side admin UX** (preview split-pane, custom React field components, publishing-checklist banner, lead-list CSV export) deferred — depends on either **apps/web** for the preview iframe or on the **leads** collection runtime which lands in Phase E.

Phase A · Bootstrap

Goal: empty repo → working Payload admin running locally and on the droplet, with CI green.

#	Ticket	Arch doc anchor	Notes
A1	Initialise pnpm workspaces + Turborepo at repo root	§#repo	pnpm-workspace.yaml, turbo.json, root package.json scripts: dev / build / lint / typecheck
A2	Create packages/config with shared tsconfig + Biome config	§#stack	Strict TS; Biome for lint+format
A3	Create packages/types skeleton (re-exports apps/cms/payload-types)	§#stack	Empty until A6 lands
A4	Scaffold apps/cms Next.js 16.2 app	§#stack	next@16.2, react@19, App Router
A5	Install Payload 3.81+, configure payload.config.ts minimal	§#stack	Postgres adapter, graphql: { disable: true }, secret from env
A6	Wire payload generate:types into pnpm script + pre-commit	§#decisions (RBAC shape row)	Output committed to apps/cms/payload-types.ts
A7	Local dev: docker-compose with Postgres 16	§#hosting	infra/docker-compose.yml; localhost-bound
A8	Auth + RBAC base:users collection with roles: string[]	§#rbac, §#decisions (RBAC row)	One bootstrap admin seeded via migration

#	Ticket	Arch doc anchor	Notes
A9	Mandatory 2FA on admin login DEFERRED to Phase I hardening — admin sits behind Cloudflare WAF + strong-password meanwhile	§#decisions (admin access row)	Plugin choice still open: payload-2fa-plugin vs custom auth strategy with TOTP
A10	Provision droplet (DO Bangalore, 2 GB / 1 vCPU + 2 GB swap)	§#hosting, §#droplet-tuning	Apply tuning spec verbatim
A11	Coolify install on droplet, repo connected, auto-deploy on main push	§#hosting	Pinned Coolify version
A12	Caddy reverse proxy in front of admin (admin.cleantart.com) with auto-SSL	§#hosting	infra/caddy/Caddyfile
A13	Cloudflare in front of admin, WAF rules enabled	§#decisions (admin access row)	Bot fight mode + rate limit on /api/users/login
A14	GitHub Actions: lint + typecheck + build on PR	§#stack	One workflow file; matrix scoped to apps/cms for now
A15	Sentry wired (frontend + Payload server)	§#stack	DSN from env; PII redaction config

Phase A exit criteria:

- **pnpm install && pnpm dev** brings up admin locally at **http://localhost:3000/admin**
- Bootstrap admin can log in with 2FA enabled
- **https://admin.cleantart.com/admin** reachable, served via Cloudflare → Caddy → Payload, with valid SSL
- Push to **main** triggers Coolify deploy and the new container takes over without admin downtime > 60s
- CI green on a no-op PR; failing lint/typecheck/build fails CI as designed

Phase B • Core schema

Goal: every collection in arch doc §03 exists with full field schema and generated types.

Build in dependency order (matches arch doc Phase 2 of migration):

#	Ticket	Arch doc anchor	Notes
B1	media collection · R2 cloud-storage adapter · Sharp transforms	§#new-fields (media)	Humanised-filename alt-text starter on upload (§#decisions alt-text row)
B2	Upload size limits per file type	§#upload-limits	Enforced server-side, surfaced in admin error
B3	users extension: roles enum, RBAC rules	§#rbac	Access control functions in apps/cms/src/payload/access/
B4	authors collection (pure content, no linkedUser)	§#new-fields (authors)	Per locked-this-session decision
B5	categories , newsCategories , jobLocations	§#new-fields	Direct 1:1 from Webflow shape
B6	forms collection: 6 field types + conditional logic + rate-limit/retention defaults	§#forms	Day-1 conditional logic per locked decision
B7	blogs collection with full field set + drafts/versions	§#new-fields (blogs)	Drafts + versions + scheduled publish
B8	news collection · isAccessibleForFree: true · NewsArticle JSON-LD	§#new-fields (news)	News-inclusion posture per locked decision
B9	guides collection · faqs[] array · keywords · articleSections · citations	§#new-fields (guides)	Replaces flattened Q1–Q5/Ans1–Ans5
B10	resources collection with optional gateForm relationship	§#new-fields (resources)	Per locked-this-session decision
B11	events + webinars with registrationMode discriminator	§#new-fields (events, webinars)	Per locked decision: inHouse / external switchable per record
B12	jobs collection · source discriminator (CMS vs ATS)	§#new-fields (jobs)	JobPosting JSON-LD layer 1
B13	aboutGalleries collection (no per-item URL)	§#not-migrating	Read-only on /about-us only
B14	pages collection (page-builder host, blocks added in Phase C)	§#new-fields	Empty layout blocks array for now

#	Ticket	Arch doc anchor	Notes
		(pages)	
B15	<code>redirects</code> collection · slug-change <code>beforeChange</code> hook	<code>§#new-fields</code> (redirects)	Hook lands in Phase D — schema only here
B16	Globals: <code>siteSettings</code> , <code>mainNav</code> , <code>footerNav</code> , <code>legal</code> , <code>seoDefaults</code> , <code>announcements</code>	<code>§#globals</code>	Singletons
B17	Reusable SEO field group <code>apps/cms/src/payload/fields/seo.ts</code>	<code>§#seo-group</code>	Wired into every content collection
B18	<code>payload generate:types</code> runs in CI; build fails on type drift	<code>§#stack</code>	Snapshot test on public type surface

Phase B exit criteria:

- Every collection in arch doc §03 + every global in `§#globals` exists in `apps/cms/src/payload/`
- Generated `payload-types.ts` compiles; `packages/types` re-exports cleanly
- An editor can create / save draft / publish one document per content collection in admin
- Drafts and scheduled publish surfaces are visible per collection
- RBAC rules enforce: admin = full, editor = create/edit/publish content, author = create/edit own drafts, never publish
- Schema-validation snapshot tests pass

Phase C · Page-builder blocks

Goal: 17 blocks composed onto `pages.layout`; admin can build a page using all of them.

#	Ticket	Arch doc anchor
C1	Lexical config with table extension (<code>colspan/rowspan</code> handling)	<code>§#table-handling</code>
C2	Typed-link picker <code>{kind: 'doc' 'media' 'url', ...}</code>	<code>§#link-handling</code>
C3	FAQ array field UX	<code>§#faq-handling</code>
C4	Block: Hero	<code>§#blocks</code>
C5	Block: FeatureGrid	<code>§#blocks</code>
C6	Block: LogoCloud	<code>§#blocks</code>
C7	Block: Testimonial	<code>§#blocks</code>
C8	Block: CTA	<code>§#blocks</code>
C9	Block: Comparison	<code>§#blocks</code>

#	Ticket	Arch doc anchor
C10	Block: FAQ (uses C3)	§#faq-handling
C11	Block: Stats	§#blocks
C12	Block: Gallery	§#blocks
C13	Block: CodeBlock	§#blocks
C14	Block: IntegrationLogos	§#blocks
C15	Block: MetricsBar	§#blocks
C16	Block: Embed	§#blocks
C17	Block: RichText (uses C1)	§#semantic-tags
C18	Block: Pricing	§#blocks
C19	Block: JobsList	§#blocks
C20	Block: Table (uses C1's table extension)	§#table-handling
C21	Wire all blocks into pages.layout	§#new-fields (pages)
C22	JSON-LD layer 2: HowTo, SoftwareApplication, VideoObject, Review (typed-field driven)	§#structured-data

Phase C exit criteria:

- Admin can compose a page using all 17 block types
- Generated types include every block discriminator
- JSON-LD preview field renders without errors for a sample page using each block
- [schema-dts](#) validates emitted JSON-LD with zero errors

Phase D • Editor experience

Goal: editors can preview, follow the publishing checklist, and ship without breaking links.

#	Ticket	Arch doc anchor
D1	Preview API: token-gated URL with JWT exp (1h/1d/7d/30d)	§#preview-workflow
D2	Admin split-pane preview UX	§#preview-workflow
D3	Slug-change beforeChange hook → auto-creates redirects row	§#migration (Route segments subsection)
D4	Publishing checklist UI: required fields, internal-link integrity, alt-text, JSON-LD validity	§#publishing-checklist
D5	Custom field component: SEO panel	§#seo-group

#	Ticket	Arch doc anchor
D6	Custom field component: FAQ array UX	§#faq-handling
D7	Custom field component: JSON-LD preview	§#structured-data
D8	Custom field component: publishing-checklist banner	§#publishing-checklist
D9	Author offboarding flow	§#author-offboarding
D10	Drafts/versions/scheduled-publish admin UX polish	§#drafts-versions
D11	Lead-list view: filters, sort, CSV export	§#forms

Phase D exit criteria:

- Publishing checklist blocks Publish on rule violations; bypass is impossible from the UI
- Slug change on a published doc creates a **redirects** row with the old → new mapping
- Preview URL works for a draft; expiry returns 401 after JWT expires; revocation works at expiry granularity (no manual revoke at v1 per arch doc)
- CSV export of leads works for a filtered list

Phase E • Forms + leads

Goal: forms render, leads land, no submission lost during a CMS outage.

#	Ticket	Arch doc anchor
E1	leads collection: append-only, PII-aware redaction in Sentry	§#privacy-gdpr
E2	LeadHandler adapter: primary writes to DB; secondary handlers fan out	§#forms
E3	Brevo handler (transactional email on submission)	§#forms
E4	Teams webhook handler (Standard Webhooks signing)	§#webhooks
E5	R2 fallback queue: /api/leads writes to R2 if DB write fails	§#forms
E6	Cron: 5-min lead-queue drain	§#cron-jobs
E7	Turnstile verification on /api/leads (no reCAPTCHA)	§#forms
E8	Form rate-limit + retention defaults applied globally	§#rate-limiting, §#privacy-gdpr
E9	E2E test: submission with DB intentionally down → R2 queue → drain on recovery	§#forms

Phase E exit criteria:

- Form submission with CMS up: **leads** row created + Brevo email fired + Teams webhook delivered with valid signature

- Form submission with DB injected-down: lead lands in R2 queue; cron drains it within 5 min; nothing lost
- Turnstile blocks a known-bad token (e2e fixture)
- Rate limit fires after configured threshold (e2e fixture)
- CSV export from D11 includes the test leads

Phase F · Search + structured data

Goal: published content is searchable; JSON-LD is valid for every collection.

#	Ticket	Arch doc anchor
F1	Meilisearch on droplet (docker-compose)	§#stack
F2	Search index schema	§#search-index
F3	Indexing via Payload afterChange hooks (publish + delete)	§#search-index
F4	Search analytics + content-gap signals	§#search-analytics
F5	404 monitoring + broken-inbound-link surfacing	§#404-monitoring
F6	JSON-LD layer 1 emission (auto from typed fields) per collection	§#structured-data
F7	/sitemap-news.xml route handler on apps/cms	§#sitemap-robots
F8	Site-level NewsMediaOrganization schema	§#decisions (news inclusion row)

Phase F exit criteria:

- Publishing a doc indexes it in Meili within 5s; deletion removes it
- Sample query returns expected typo-tolerant + faceted results
- **schema-dts** validates JSON-LD output for one fixture per content collection with zero errors
- **/sitemap-news.xml** returns valid news sitemap with **isAccessibleForFree: true** per item

Phase G · Webhooks, cron, observability

Goal: every operational signal arch doc requires is wired and verifiably firing.

#	Ticket	Arch doc anchor
G1	Webhook emitter: document.published + lead.submitted to Teams	§#webhooks
G2	Standard Webhooks signing on emit; verification fixture	§#webhooks

#	Ticket	Arch doc anchor
G3	Cron catalog: backup heartbeat, lead-queue drain (E6), Meili reindex check, IndexNow ping, sitemap regen, scheduled-publish runner	§#cron-jobs
G4	Structured JSON logging to stdout · request-ID propagation	§#logging-alerting
G5	Sentry PII redaction policy in code	§#privacy-gdpr
G6	BetterStack Uptime: 5 monitors (admin, Meili, SSL, backup heartbeat, droplet ping)	§#logging-alerting
G7	Alert rules: 10 rules per arch doc; P1 pages, P2 Slack, P3 daily digest	§#logging-alerting
G8	Status page at status.cleanstart.com	§#decisions (uptime monitoring row)

Phase G exit criteria:

- Publishing a doc fires Teams webhook with valid signature
- Backup-heartbeat-missing alert fires when cron is paused for 25h in a test env
- Sentry captures a thrown error from a hook with PII redacted
- Structured log line shows request-ID end-to-end through one publish action

Phase H · Migration ETL

Goal: dry-run on staging passes all 9 acceptance criteria from arch doc [§#migration](#).

#	Ticket	Arch doc anchor
H1	migrations/webflow-import/export.ts : Webflow API → JSONL · token-bucket throttle 50 req/min	§#migration (Phase 1)
H2	Schema-parity audit script → migrations/SCHEMA-PARITY-AUDIT.md	§#migration (Phase 1)
H3	Per-collection transforms: transform-{collection}.ts	§#migration (Phase 2)
H4	Body-text URL rewriter (uploads-ssl.webflow.com / uploads.webflow.com → R2)	§#migration (zero-tolerance subsection)
H5	Asset re-upload with migration_progress checkpoint table + SHA-256 verify	§#migration (Asset re-upload subsection)
H6	import.ts : Payload Local API loader, dependency-ordered	§#migration (Phase 2)
H7	url-parity.ts : sitemap diff Webflow vs new	§#migration
H8	Acceptance-criteria gate: 9 automated assertions	§#migration (Acceptance criteria table)

#	Ticket	Arch doc anchor
H9	Staging dry-run runner	§#migration (Phase 3)
H10	Rollback drill mechanics	§#migration (Rollback procedure)

Phase H exit criteria:

- Dry-run on staging passes all 9 acceptance criteria
- Rollback drill executed at least once on staging
- Sign-off captured in `docs/migration/dry-run-{date}.md` per arch doc

Phase I · Hardening + cutover-ready

Goal: prod-quality posture; ready for cutover-day runbook.

#	Ticket	Arch doc anchor
I1	Security headers (CSP, HSTS, etc.)	§#security-headers
I2	Rate limiting on public endpoints	§#rate-limiting
I3	<code>infra/scripts/backup.sh</code> + <code>restore.sh</code> · cron + heartbeat	§#restore-runbook
I4	Quarterly restore drill scheduled · <code>docs/RESTORE-LOG.md</code> started	§#decisions (restore drill row)
I5	Droplet tuning spec applied (Postgres / Sharp / Meili / kernel)	§#droplet-tuning
I6	Admin Lighthouse run · p95 admin action < 1s benchmark	§#perf-budget
I7	Pre-cutover smoke test suite (50 sample-URL HTTP-200 + admin login + form submit)	§#migration (Phase 4 T-15m)

Phase I exit criteria:

- Restore drill: yesterday's backup restored to a fresh droplet within documented RTO
- Security-headers scan (e.g. securityheaders.com) returns A+ for `admin.cleanstart.com`
- Admin Lighthouse score above target on 5 representative routes
- Smoke test suite green on staging

Cross-cutting CI gates (apply on every PR after Phase A)

- `pnpm --filter @cleanstart/cms lint`
- `pnpm --filter @cleanstart/cms typecheck`

- `pnpm --filter @cleanstart/cms build`
 - `pnpm --filter @cleanstart/cms test` (Vitest)
 - `pnpm --filter @cleanstart/cms e2e --grep @phase-{current}` (Playwright; phase-tagged)
 - `payload generate:types` drift check
 - `schema-dts` validation on JSON-LD fixtures (Phase F onward)
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Out of scope this backlog

- `apps/web` (public marketing site) — separate backlog after public-site planning session
- Brand/Tailwind tokens, shadcn theming — wait for new brand + Figma
- Custom `/dashboard` route group — only added if Payload admin proves insufficient (per arch doc §`#decisions` custom-dashboard row)
- ISR-revalidate handshake consumer side — lives in `apps/web`
- Public-site sitemap.xml — lives in `apps/web`